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Do body size mean and variance share the same genetic architecture?

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Abstract

Individuals sharing the same genotype often differ in their phenotypes for quantitative traits, even when reared under common environmental conditions. This phenomenon, known as intra-genotypic variance, represents an important but poorly understood component of phenotypic variation. Here, we investigated whether body size mean and intra-genotypic variance share the same genetic architecture in the inbred, sequenced lines of the *Caenorhabditis elegans* multiparental experimental evolution (CeMEE) panel. Body size was measured under two environmental conditions: a standard laboratory environment and a stressful high-salt environment. We quantified genetic variation in both trait mean and intra-genotypic variance, estimated broad-sense heritability, and performed genome-wide association studies (GWAS) to identify loci associated with each trait component. We found substantial genetic variation for both mean and variance. Broad-sense heritability estimates for variance traits were comparable to, and in some cases exceeded, those of the mean. Intra-genotypic variance differed markedly between environments, indicating a strong plastic component. GWAS identified loci associated with both body size mean and phenotypic variance, with only partial overlap between their genetic architectures. While several loci influenced both components, others were specific to either mean or variance, suggesting that intra-genotypic variance can evolve partly independently from trait means.

Contents

Abstract **1**

List of abbreviations **3**

Introduction **4**

Materials and methods **7**

CeMEE panel 7

Environmental conditions 7

Body size measurements 8

Statistical models for mean and intra-genotypic variance 8

GWA analyses 10

Results **11**

Correlation of mean and intra-genotypic variance 12

Distribution of mean and intra-genotypic variance 13

Heritability for intra-genotypic mean and variance 15

Genetic variation for intra-genotypic mean and variance 16

Genetic variation for intra-genotypic mean and variance is environment-specific 16

Discussion **20**

Conclusion **23**

List of abbreviations

- NGM = Nematode Growth Medium
- vQTL = variance Quantitative Trait Loci
- GWAS = Genome-Wide Association Studies
- *CeMEE* = *Caenorhabditis elegans* Multiparental Experimental Evolution
- CV = Coefficient of Variation
- FDR = False Discovery Rate
- LD = Linkage Disequilibrium

Introduction

Phenotypic variation within populations arises from two main sources: genetic differences between individuals and environmentally induced effects. Traditionally, evolutionary studies have focused on inter-genotypic variation, that is, differences in mean trait values between genotypes. Such differences provide the primary substrate for natural selection and underpin most quantitative genetic analyses.

However, phenotypic diversity cannot be fully understood by considering variation in trait means alone. Substantial variation may also arise within genotypes. In constant environments, genetically identical individuals are often assumed to produce uniform phenotypes. However, empirical evidence questions this assumption: phenotypic variance is consistently observed within isogenic populations maintained under controlled conditions [1]. This intra-genotypic variance reveals that genetically homogeneous backgrounds can generate heterogeneous phenotypic outcomes.

Intra-genotypic variance is commonly attributed to the existence of stochastic micro-environmental variation that affects development, and the ability of an individual's genotype to buffer development against these micro-environmental perturbations, as each organism effectively experiences a unique local environment. These variations arise because individuals sharing the same genotype can differ in their phenotypes due to small fluctuations in the developmental, physical, and social environment to which individuals with the same genotype are exposed. These non-genetic sources of variation can generate substantial dispersion in trait expression, even in populations lacking genetic diversity. One adaptive explanation of the phenomenon of intra-genotypic variance in isogenic populations is the evolution of bet-hedging strategy, which makes the link between environmental variability, fitness variance and mean fitness [2] [3]. Bet-hedging strategies are those that minimize fitness variation across generations in the face of environmental variation. Thus, when comparing phenotypically invariant and variant genotypes, an invariant genotype would have a certain fitness in one environment to which it was adapted, but a potentially steep fitness decline outside of that environment. In contrast, a phenotypically variant genotype could have a comparatively greater fitness in heterogeneous environments because its fitness would not decline so quickly as the environment changes. Therefore, intra-genotypic variance increases the fit phenotypic space of a genotype.

The variance observed among genetically identical individuals was often treated as residual

noise around a genetically determined mean. However, theoretical work suggested that genetic factors might also influence the variance of phenotypes itself. As early as the 1980s, quantitative genetic models proposed that loci could affect the variance of a trait independently of its mean [4]. Under this view, intra-genotypic variance is not merely a product of environmental noise, but may instead represent a biologically meaningful property that can be under genetic control.

Empirical support for this idea has emerged from studies identifying loci that influence intra-genotypic variance. Such loci, commonly referred to as variance quantitative trait loci (vQTL), affect the degree of variance expressed by individuals sharing the same genotype. vQTL have been identified in several systems, including yeast [5] and *Arabidopsis thaliana* [6], demonstrating that genetic variants can modulate not only the average value of a trait but also its variance. More recently, the development of genome-wide association studies (GWAS) has enabled the systematic detection of such loci on a genome-wide scale. Using these approaches, variance-associated loci have now been reported for a wide range of traits and organisms. Examples include locomotor handedness and bristle number in *Drosophila* [7][8], flowering time in *Arabidopsis thaliana* [6][9], morphological traits in maize (*Zea mays*) [10], calcium excretion in rats (*Rattus norvegicus*) [11], litter size in pigs (*Sus scrofa*) [12] and body mass index in humans [13]. Variance-control loci are of particular interest because they may reflect fundamental biological processes governing the stability and robustness of phenotypes. Genetic effects on intra-genotypic variance have been associated with mechanisms such as micro-environmental plasticity [14][15], canalization which is defined as the robustness of phenotypes to perturbation [16][17], developmental stability [18] and natural variation in stochastic developmental noise [6].

Despite these advances, it is still unclear how common variance-controlling loci are in genomes and how much they contribute to the overall variation of traits in natural populations. Understanding the genetic basis of phenotypic mean and intra-genotypic variance is therefore essential for elucidating the mechanisms that generate, buffer, or amplify variance, and for understanding how these processes influence evolutionary dynamics.

To study the genetic architecture of mean and intra-genotypic variance, we used a panel of wild-derived *Caenorhabditis elegans* inbred lines (CeMEE panel) domesticated under laboratory conditions. Inbred line designs are powerful for estimating the genetic basis of intra-genotypic variance because all variance between individuals within a completely homozygous inbred line is

due to intra-genotypic variance. For each line, this allows us to measure any phenotype in a large number of individuals of the same genetic background, age, and rearing environment, thus empirically estimating the magnitude of mean and intra-genotypic variance. Specifically, we measured the body size of *C. elegans*.

Only one article focuses on intra-genotypic variance in *C. elegans*, by studying isogenic lines in two environments, and suggests that the variance in lifetime fecundity may be a bet-hedging strategy, which is a strategy increasing intra-genotypic variance to cope with unpredictable environments [19].

Given the complexity of these interacting regulatory mechanisms, body size is expected to be a highly polygenic trait [20], controlled by numerous loci of small effect. Accordingly, we expect genome-wide association analyses to identify multiple SNPs contributing to variation in mean body size across genotypes. In contrast, under standard laboratory conditions, an environment to which *C. elegans* strains have been maintained for many generations, intra-genotypic variance may be constrained by stabilizing selection favoring robust and canalized developmental outcomes. Under such conditions, the genetic architecture underlying intra-genotypic variance itself may therefore be limited, resulting in fewer detectable loci associated with intra-genotypic variance. Conversely, under stressful conditions, cryptic genetic variation may be revealed, and genetic variants influencing intra-genotypic variance may become detectable. Such variance-modulating loci could contribute to differential adaptive potential by increasing developmental flexibility in response to environmental stresses, and could be involved in a bet-hedging strategy, whereby increased phenotypic variability among individuals sharing the same genotype enhances fitness under unpredictable environmental conditions.

We have therefore examined the body size of *C. elegans* and, more particularly, how this varies among individuals. Specifically, we asked for an inbred line of *C. elegans* in one environment how much intra-genotypic variance in body size there was among individuals. We compared inbreed lines of *C. elegans* and found that the among individual variation in body size within each line varied. We then repeated this but using different environments, one in low-salt concentration and one in a stressful environment with high-salt concentration, which tests how the environment itself affects this intra-genotypic variance. We also compared two metrics of intra-genotypic variance: variance corrected for the mean and the coefficient of variation (CV). Comparing these two metrics

allowed us to assess whether patterns of intra-genotypic variance were consistent across different measures of variance.

Materials and methods

CeMEE panel

We used the *Caenorhabditis elegans* Multiparental Experimental Evolution (CeMEE) panel, a genetically diverse population derived from multiple generations of experimental evolution. It is a multiparental recombinant inbred line (RIL) developed to dissect the genetic architecture of complex traits with high mapping resolution. The CeMEE panel was founded by intercrossing 16 wild *C. elegans* isolates originating from various geographic locations, thus capturing a substantial fraction of the standing natural genetic variation. A6140 derives from the hybridization of the 16 founder strains over 33 generations, followed by 140 generations of evolution under a standardized laboratory environment. In this nomenclature, the prefix "A" denotes an androdioecious mating system, the digit "6" identifies the replicate population, and "140" indicates the number of generations elapsed since domestication under laboratory conditions [21].

Additionally, the *C. elegans* lines analyzed in this study originated from distinct experimental evolution regimes derived from the ancestral A6140 population. These populations were maintained under different selective and environmental conditions, generating genetically differentiated lineages adapted to contrasting evolutionary contexts [22][23]. Together, these populations constitute a genetically diverse framework for investigating the genetic architecture of body size mean and variance.

The genome of *C. elegans* is composed of five autosomes (chromosomes I–V) and one sex chromosome (X), with hermaphrodites carrying two X chromosomes (XX) and males carrying a single X chromosome (XO). Inbreed lines only contain hermaphrodites.

Environmental conditions

Lines have been maintained under standard laboratory conditions in 9 cm Petri dishes with Nematode Growth Medium (NGM) and *E. coli*. This bacteria serves as food for juveniles (stage L1) until reproduction. Each Petri dish contains 1000 individuals, which are reproduced during days 2 and

3, and embryos are collected on day 3 and synchronized to stage L1 for the next generation, which allow discrete non overlapping generations.

Same lines experienced a stressful environment with the same environmental conditions as before and a high concentration of NaCl (305 mM). This allows for a comparison of the genetic architecture of worms depending on the environment.

We refer to the 305 mM NaCl environment as the "high-salt" environment, while the domestication 25 mM NaCl environment as the "low-salt" environment.

Body size measurements

For this study, we used body size measurement with the Multi-Worm Tracker version 1.3.0 [24], with 4718 movies (2613 for the low-salt environment and 2105 for the high-salt environment). Each movie contains tracks of approximately 1000 worms, and the movement of tracked-object per movie was assigned to a movement state. Movie frames were sampled only for forward tracked-objects, and the body size was estimated by the number of pixels (pixel^2). For each worm, several body size measurements were obtained across time and summarized by their median value. The mean body size was then calculated at the plate level using these individual median values, and intra-genotypic variance was estimated between individuals within each plate.

Temperature (°C), relative humidity at the time of video recording, and the logarithm of object density within each Petri dish (which represent the number of individuals) were measured. Session and block were quantified as grouping factors corresponding to experimental batches and temporal sampling structure. For each line, we also have multiple replicate.

Statistical models for mean and intra-genotypic variance

To characterize both trait expression and intra-genotypic variance, we analyzed body size mean, variance, and coefficient of variation (CV). CV was included to quantify relative variance independently of differences in mean trait values, thereby reducing the intrinsic scaling relationship between the mean and variance, defined as the standard deviation divided by the mean:

$$CV = \frac{\sigma}{\bar{y}} \times 100$$

We removed six measurements from the low-salt environment and eight measurements from the high-salt condition whose values exceeded three standard deviations from the mean of the variance.

To account for environmental covariate during the experiments, the mean and CV of body size were fitted as multivariate response variables in linear mixed-effects models implemented in R (version 2024.09.1+394). Temperature, relative humidity, and the logarithm of object density within each Petri dish were included as fixed effects, while session, block, and line identity were modeled as random effects. Intra-genotypic variance was fitted using the same model structure, with body size mean included as a covariate to account for the correlation between mean and variance.

These models generated line-specific phenotypic estimates corrected for environmental covariates and experimental structure, which are used for analysis of phenotypic values.

Mean body size and CV were analyzed using linear mixed-effects models of the general form:

$$y = \mu + \alpha \times [T, RH, \log D] + \beta + \gamma + \zeta + \varepsilon$$

where μ represents the intercept, and α corresponds to the fixed environmental effects of temperature (T), relative humidity (RH), and log density ($\log D$). The term β denotes the line identity. The random effects $\gamma \sim N(0, \sigma^2)$ and $\zeta \sim N(0, \sigma^2)$ account for variation associated with assay year (session) and experimental block identity, respectively. Finally, $\varepsilon \sim N(0, \sigma^2)$ represents the residual error.

For variance analyses, mean body size was additionally included as a fixed covariate:

$$y_{\text{var}} = \mu + \text{Mean} + \alpha \times [T, RH, \log D] + \beta + \gamma + \zeta + \varepsilon$$

Variance and CV were log-transformed prior to analyses to follow a normal distribution. Because multiple biological replicates were available for each line, adjusted line-specific phenotypic estimates were subsequently extracted using the `emmeans` package [25], allowing the estimation of a single corrected phenotypic value per line after accounting for environmental covariates and experimental structure. In total, corrected phenotypic estimates were obtained for 1091 lines in the low-salt environment and 1034 lines in the high-salt environment.

For the correlations values, we computed the Pearson product moment correlations.

To test for significant differences in body size mean, variance, and CV between environments, we extended the linear mixed-effects models described above by explicitly including environment as a fixed effect and by testing for environment-specific phenotypic effects. We assessed the significance using type-II ANOVA with the `Anova` function from the car package [26]. All models were performed using `lme4` package [27].

GWA analyses

Genome-wide association studies (GWAS) were performed using `GridLMM` [28] in R, which implements linear mixed models for genome-wide association mapping while accounting for population structure and relatedness. For each phenotype, association analyses were conducted using the same modeling framework described above, including the environmental covariates temperature, relative humidity, and log density, as well as the date and session identity. In contrast to the estimation procedure described previously, line identity was here fitted as a random effect rather than a fixed effect, allowing the partitioning of genetic and residual sources of variation during marker association testing. In total, 966 lines were analyzed for this GWAS.

We considered a set of sequenced lines, which had been sequenced during previous experiments at the lab [23][29], and some were sequenced at a New York laboratory. All lines were already mapped to the reference genome WS283. Variants were filtered to retain only polymorphic sites with a minor allele frequency (MAF) exceeding 5%, thereby excluding rare variants and non-polymorphic positions from subsequent analyses. A total of 317,611 SNPs were included in the analyses.

To reduce redundancy due to linkage disequilibrium among markers, we performed SNP pruning prior to downstream analyses. Pruning was conducted using PLINK v1.9, applying a sliding window approach of 200 SNPs, a step size of 10 SNPs, and an r^2 threshold of 0.8 (`--indep-pairwise` 200 10 0.8). This procedure was applied to the imputed VCF dataset, resulting in a subset of approximately independent markers. The list of retained SNPs was extracted from the resulting `.prune.in` file and used for subsequent analyses. This step allowed us to mitigate the effects of linkage disequilibrium and reduce inflation of association signals driven by correlated variants.

For the same reason as before, SNPs were LD-pruned prior to kinship estimation using PLINK,

ensuring that relatedness was not disproportionately driven by correlated markers. Variants with an $r^2 > 0.1$ were removed using the `-indep-pairwise` procedure with 200-kbp windows and a sliding step of 10 variants. The resulting LD-pruned dataset was then used to construct chromosome-specific genomic relationship matrices using the VanRaden method [30].

To avoid proximal contamination, association analyzes were performed using a leave-one-chromosome-out (LOCO) framework [31]. For each focal chromosome, the kinship matrix was computed from markers on all other chromosomes, excluding the chromosome being tested. This prevents the tested SNPs from contributing to the kinship correction used in their own association test, thereby reducing bias in effect-size estimates while maintaining appropriate control for genome-wide population structure.

To correct for multiple testing, *P*-values obtained from the `GridLMM` models were adjusted using the Benjamini–Hochberg false discovery rate (FDR) procedure [32], implemented with the `p.adjust` function in R. SNPs with FDR-adjusted *P*-values below the significance threshold of 0.05 were considered statistically significant.

Broad-sense heritability (H^2) was estimated from variance components obtained in the previous model in `GridLMM` by fitting line identity as a random effect. This approach allowed us to partition total intra-genotypic variance into a line-specific component, representing genetic variance among lines. The resulting heritability estimate was interpreted as the fraction of intra-genotypic variance explained by genetically distinct lines. Because GWAS were performed separately for each chromosome, multiple H^2 estimates were obtained, one for each chromosome. Mean heritability was then calculated across chromosomes for the mean, variance, and CV of body size, in each of the two environments.

Data, R code, and modeling results are available in the GitHub repository.

Results

We assessed the suitability of two metrics and their natural logarithms for quantifying intra-genotypic variance: the within-line variance (σ^2) and the within-line coefficient of variation (CV). We estimated these metrics for one quantitative trait: body size. $\ln(\sigma^2)$ and ln(CV) were nearly normally distributed for all analyzes; the non-transform metrics were highly skewed. The correlations be-

tween these metrics are high with r = 0.727 ($P < 0.01$) for the low-salt environment and r = 0.934 ($P = 0$) for the high-salt environment (Fig. 1a).

Correlation of mean and intra-genotypic variance

Trait means and variances are often intrinsically correlated, such that lines with larger mean values tend to exhibit larger variances. This mean–variance relationship can confound the interpretation of genetic effects on intra-genotypic variance if not properly accounted for. To assess whether our modeling approach successfully removed this dependency, we examined the correlation between line means and our two variance-based phenotypes: log-transformed variance $\ln(\sigma^2)$ and log-transformed coefficient of variation ln(CV).

Before correcting for this dependency and environmental covariates, the correlations between mean and variance were positive in both environments (r = 0.428 , $P < 0.01$ for the low-salt environment ; r = 0.314 , $P < 0.01$ for the high-salt environment).

After correction, our data on the correlation between line mean and variance using $\ln(\sigma^2)$ to parameterize intra-genotypic variance show a low positive correlation for the low-salt environment with r = 0.216 ($P < 0.01$), and a low negative relationship for the high-salt environment with r = -0.096 ($P < 0.01$) (Fig. 1b).This indicates that the association between mean and variance persists after accounting for environmental and experimental sources of variation.

CV is often used to remove any relationship between mean and variance. However, using ln(CV) to parameterize this intra-genotypic variance did not completely remove its correlation with the mean for both environments (r = -0.497 , $P < 0.01$ for the low-salt environment ; r = -0.358 , $P < 0.01$ for the high-salt environment) (Fig. 1c). Thus, while both $\ln(\sigma^2)$ and ln(CV) indicate variable relationships between trait means and estimates of variance, the magnitude and sign of the correlations vary depending on what metric is used to quantify intra-genotypic variance. Since the correlations between mean and variance are environment-specific, no single measure of variance will result in the same relationship for this trait, so that the choice of the appropriate metric will necessarily depend on the purpose of the analysis.

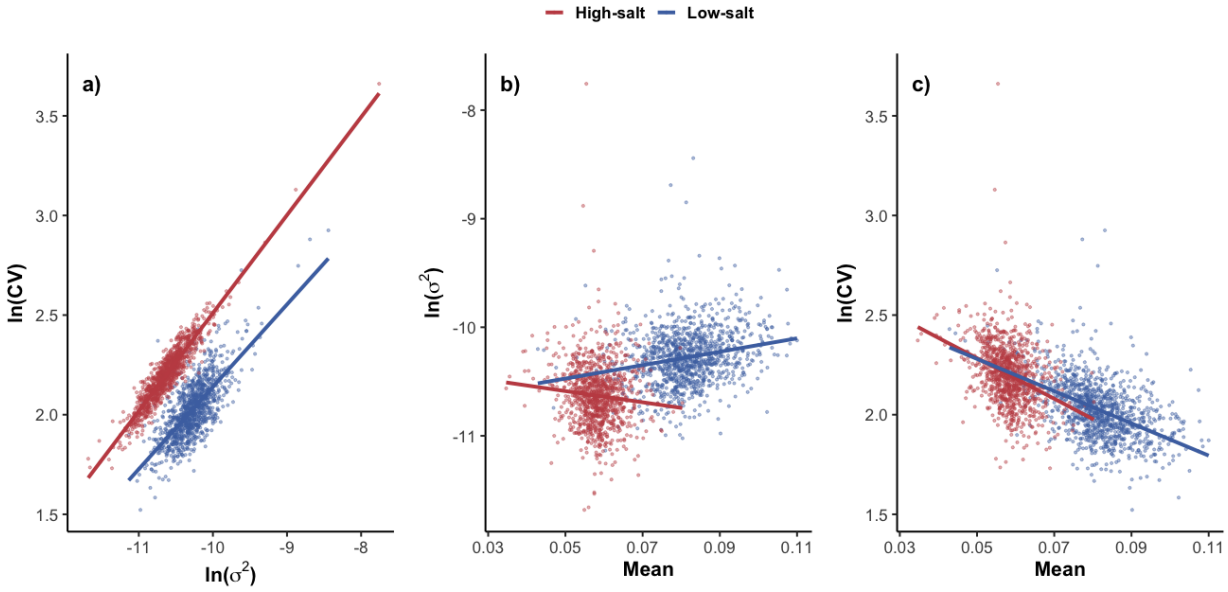


Figure 1: **Distribution of mean and variance of body size after correction for high-salt environment (red) and low-salt environment (blue).** (a) The correlations between ln(σ^2) and ln(CV). (b) The correlations between the mean and ln(σ^2). (c) The correlations between the mean and ln(CV).

Distribution of mean and intra-genotypic variance

We tested whether there was phenotypic heterogeneity among lines, and found highly significant heterogeneity of mean, ln(σ^2), and ln(CV) in both environments. In the low-salt environment, significant line effects were detected for mean body size ($\chi^2_{1090} = 3429.05$, $P < 0.01$), ln(σ^2) ($\chi^2_{1090} = 1543.59$, $P < 0.01$), and ln(CV) ($\chi^2_{1090} = 1643.24$, $P < 0.01$). Similarly, in the high-salt environment, line effects were highly significant for mean body size ($\chi^2_{1033} = 2087.16$, $P < 0.01$), ln(σ^2) ($\chi^2_{1033} = 1616.74$, $P < 0.01$), and ln(CV) ($\chi^2_{1033} = 1741.77$, $P < 0.01$). This indicating the presence of genetic variation in intra-genotypic mean and variance.

We then partitioned the variation in mean and variance into environmental components. Significant differences in mean body size were detected between environments, with smaller worms in the stressful high-salt environment ($\chi^2_1 = 10906.55$, $P < 0.01$) than in the low-salt environment (Fig. 2a). Surprisingly, we found higher ln(σ^2) for the low-salt environment ($\chi^2_1 = 308.11$, $P < 0.01$) and a lower ln(CV) for the same environment ($\chi^2_1 = 514.30$, $P < 0.01$) compared to the high-salt environment (Fig. 2b and 2c). The observed differences in intra-genotypic variance across environments indicate that phenotypic variability is not solely an intrinsic property of the genotype, but is also strongly modulated by environmental conditions.

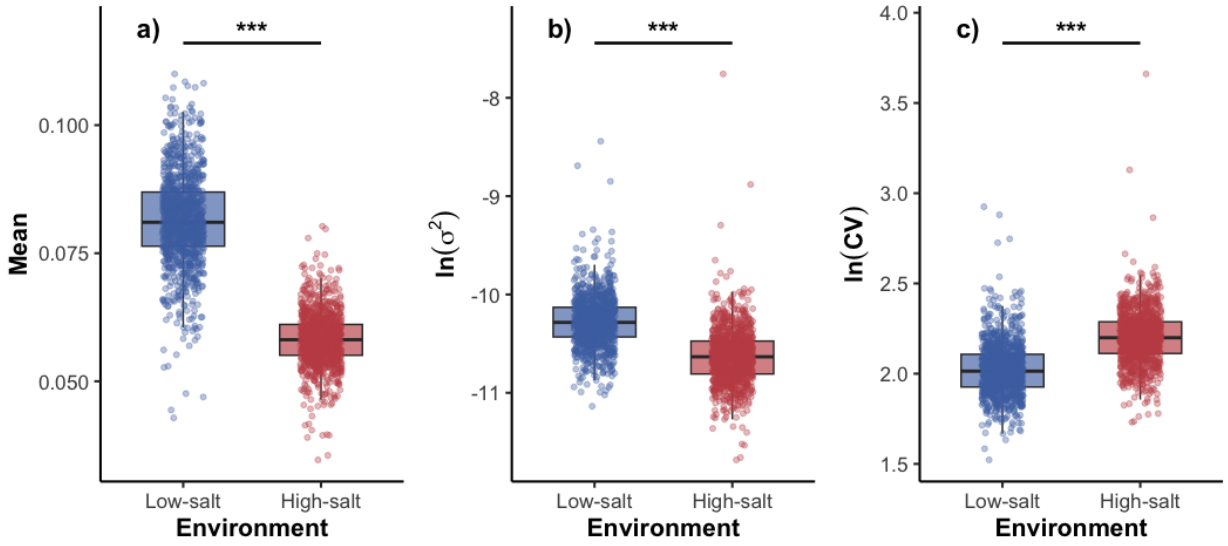


Figure 2: **Distribution of mean and variance depending on the environment.** Median and quartiles distribution of (a) phenotypic mean. (b) ln(σ^2). (c) ln(CV). Distribution are for the low-salt environment (blue) and high-salt environment (red). The box-plots represent the distribution of the data, while the circles represent lines.

Cross-environment correlations were low for both ln(σ^2) and ln(CV), indicating substantial environmental dependence in the expression of phenotypic variance (ln(σ^2): r = 0.174 , $P < 0.01$) ; ln(CV): r = 0.263 , $P < 0.01$)(Fig. 3). This mean that for one line, the intra-genotypic variance expressed in one environment is not the same amount in the other environment.

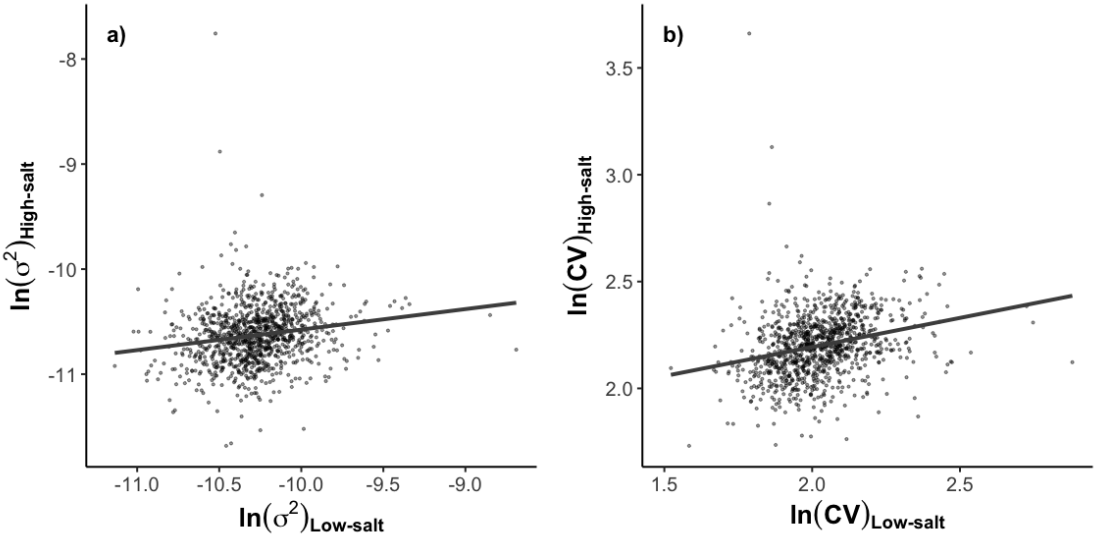


Figure 3: **Environmental consistency of intra-genotypic variance.** (a) Cross-environment correlation of ln(σ^2) between high-salt and low-salt conditions. (b) Cross-environment correlation of ln(CV) between high-salt and low-salt conditions.

Heritability for intra-genotypic mean and variance

The magnitude of the genetic variation underlying mean and intra-genotypic variance was substantial (Fig. 4). Broad-sense heritability estimates (H^2) for $\ln(\sigma^2)$ reached $H^2 = 0.12$ in the high-salt environment, compared to $H^2 = 0.03$ in the low-salt environment. By comparison, broad-sense heritabilities estimated at the level of the phenotypic mean using the same dataset were $H^2 = 0.05$ in the high-salt environment and $H^2 = 0.13$ in the low-salt environment. Thus, heritability estimates for $\ln(\sigma^2)$ were not of the same order of magnitude as those estimated for the mean phenotype. In the high-salt environment, the heritability of intra-genotypic variance was more than twice that estimated for the mean phenotype, suggesting that stressful conditions may enhance the genetic contribution to intra-genotypic variance itself.

Similar patterns were observed for ln(CV). Broad-sense heritability estimates for ln(CV) were moderate in both environments, with $H^2 = 0.06$ in the low-salt environment and $H^2 = 0.11$ in the high-salt environment. These results further support the idea that genetic control of intra-genotypic variance is maintained across environments and may become more pronounced under stressful conditions.

We can note that that broad-sense heritability of the mean is lower in the high-salt environment than in the low-salt environment, following an inverse pattern to the distribution of variance measurements (Fig. 4).

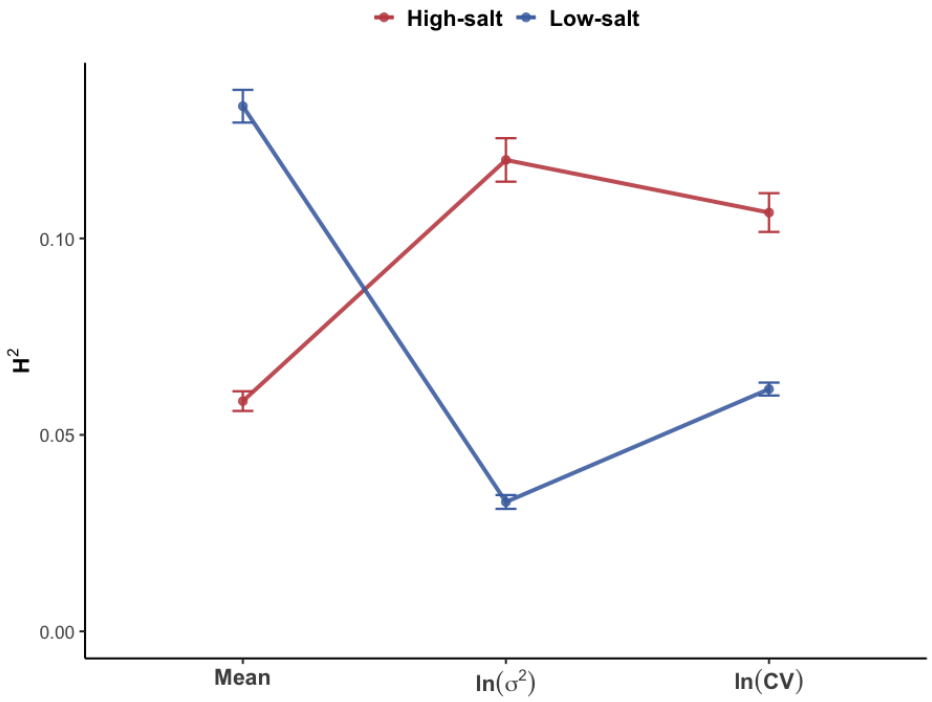


Figure 4: **Environmental variation in the broad-sense heritability of mean body size and variance.** Points represent mean broad-sense heritability estimates ($H^2 \pm$ s.d.) for mean body size, ln(σ^2), and ln(CV) in low-salt (blue) and high-salt (red) environments.

Genetic variation for intra-genotypic mean and variance

Genome wide association analysis has been done to describe the genetic architecture of the mean and intra-genotypic variance (Fig. 5).

In the low-salt environment, we found 98 significant single nucleotide polymorphisms (SNPs) at FDR < 0.05 for the mean body size. Among the five SNPs associated with ln(CV), three were also significantly associated with the mean phenotype, indicating a partial overlap between the genetic architectures of mean and intra-genotypic variance. We did not find any variants associated with ln(σ^2) (Fig. 5).

Genetic variation for intra-genotypic mean and variance is environment-specific

We tested whether the genetic basis for variance is shared between the two environments by partitioning the variance in ln(σ^2) and ln(CV) between the environment and among lines.

Interestingly, the number of SNPs associated with mean body size was four times lower in the

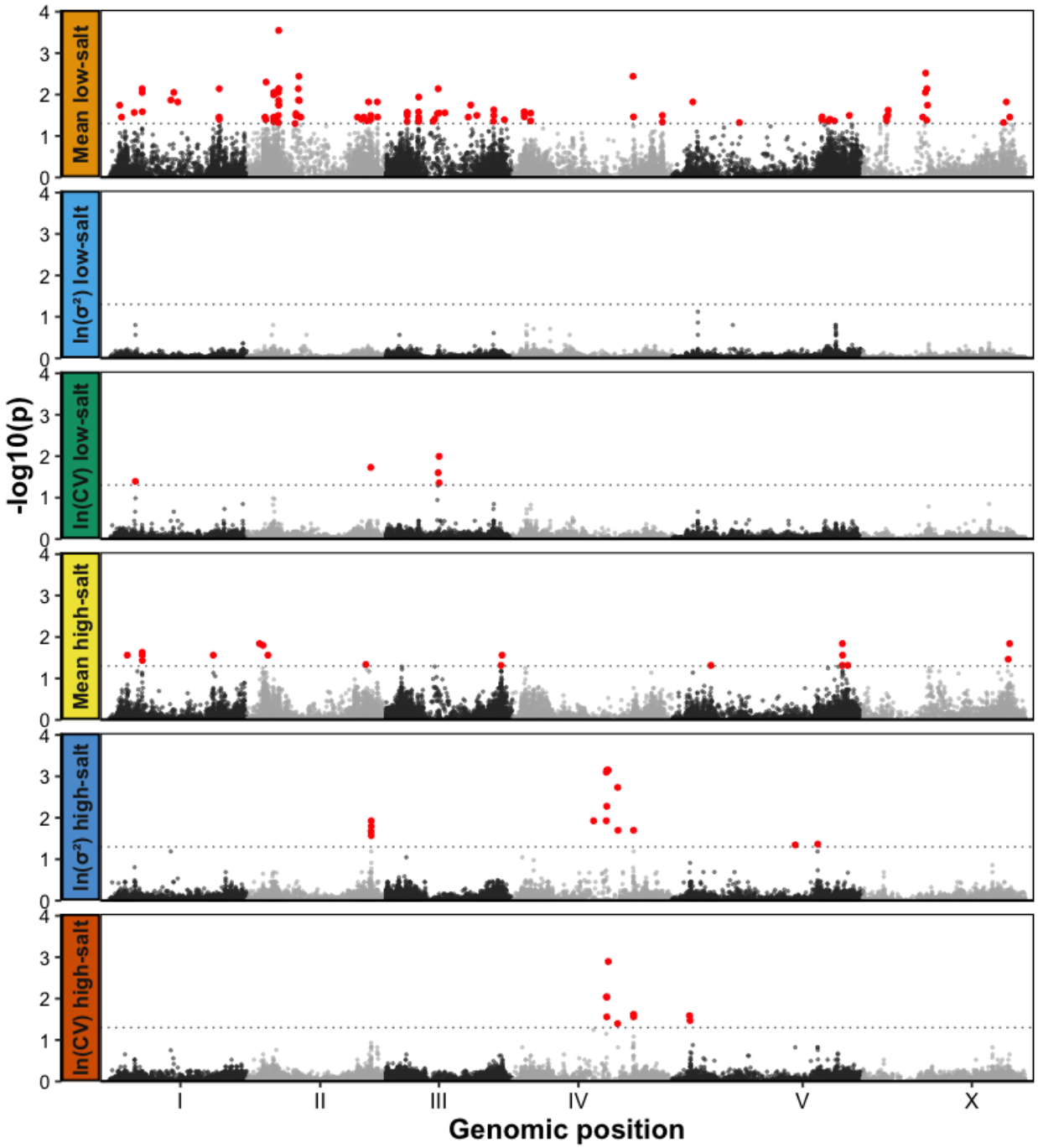


Figure 5: **Manhattan plots showing genome-wide associations for phenotypic mean and metrics of variance across the low-salt environment and high-salt environment.** Points correspond to individual SNPs ordered along chromosomes I–V and X. Each point represents a single SNP retained after linkage disequilibrium pruning ($r^2 < 0.8$). Alternating grey shades distinguish chromosomes, and colored points highlight SNPs remaining significant after Benjamini–Hochberg correction. Horizontal dashed lines indicate the significance threshold (FDR = 0.05).

high-salt environment, with only 16 SNPs detected. In addition, we identified numerous SNPs associated with intra-genotypic variance, including 15 SNPs for ln(σ^2) and 9 SNPs for ln(CV) in high-salt environment (Fig. 6). 6 SNPs are shared between both measures of intra-genotypic variance in the high-salt environment only.

We therefore infer that the genetic basis of variation in variance is environment-specific, at least for these two traits.

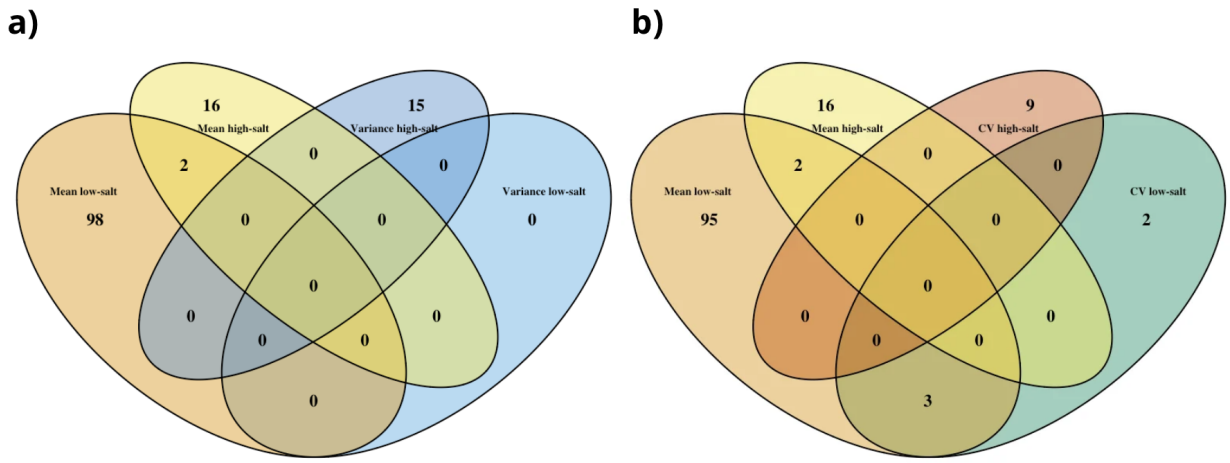


Figure 6: **Venn diagram showing the overlap of significant SNPs associated with phenotypic mean and variance across low-salt and high-salt environments.** (a) Cross-environment shared SNPs between ln(σ^2) and mean in both envrionments. (b) Cross-environment shared SNPs between ln(CV) and mean in both envrionments. Lighter orange circles correspond to SNPs associated with mean, lighter blue to ln(σ^2) and green to ln(CV) in low-salt environment. Yellow correspond to mean, blue to ln(σ^2) and orange to ln(CV) in high salt environment. Numbers indicate the number of shared or unique significant SNPs among trait–environment combinations.

Moreover, some SNPs are shared between mean body size in the two environments, with 2 SNPs that are consistently associated with mean body size in both conditions (Fig. 6). There is no share of SNPs for the metrics of intra-genotypic variance between environments.

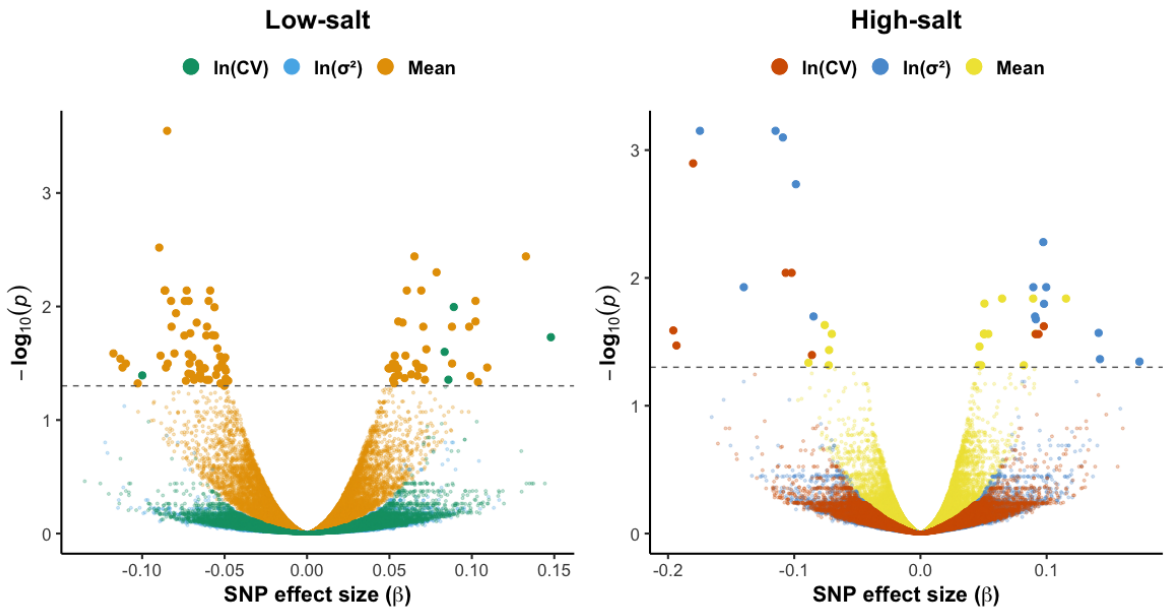


Figure 7: **Distribution of SNP effect sizes across trait means and variance-related traits under low- and high-salt environments.** Volcano plots showing the relationship between SNP effect size estimates (β) and statistical significance for GWA analyses of the mean, ln(σ^2), and ln(CV) in low-salt and high-salt environments. Each point represents a single SNP retained after linkage disequilibrium pruning ($r^2 < 0.8$). The x-axis shows the estimated additive SNP effect (β) obtained from the mixed-model GWAS, whereas the y-axis shows the FDR-adjusted significance level. Colors indicate the trait analyzed. Larger points correspond to SNPs reaching the significance threshold (FDR < 0.05), indicated by the horizontal dashed line.

A notable difference between environments emerged from the distribution of SNP effect sizes (Fig. 7). Under low-salt conditions, loci associated with body size mean generally exhibited relatively small effects, consistent with a highly polygenic architecture in which phenotypic variation is distributed across many variants of modest contribution. In contrast, loci detected under high-salt conditions tended to have larger effects, both for the mean and for measures of intra-genotypic variance. This pattern suggests that environmental stress amplifies the phenotypic consequences of genetic variation, allowing a subset of variants to contribute more strongly to trait expression. Such an increase in effect size under stress is consistent with the idea that environmental perturbations can reveal genetic effects that remain buffered or only weakly expressed under laboratory conditions.

Discussion

Correlations between mean and intra-genotypic variance

The aims of our research was to discover if intra-genotypic mean and variance share the same genetic architecture.

For that, the first step was to correct for the correlation between mean and variance. The two variance metrics, ln(CV) and ln(σ^2), were highly correlated, as expected given that both quantify intra-genotypic variance. Consistent with theoretical expectations, both metrics also exhibited a positive relationship with the phenotypic mean. Our correction procedures markedly reduced this intrinsic dependency, although a weak residual correlation remained. Such residual associations may reflect experimental or measurement-related biases. Interestingly, the strength of the mean–variance relationship differed across environments, suggesting that environmental conditions modulate the coupling between trait means and intra-genotypic variance. Variable relationships between trait means and variance are consistent with previous studies in *C. elegans* [19] and a wide range of species [15], where mean-variance correlations take on all possible values. Notably, ln(σ^2) remained less correlated with the phenotypic mean than ln(CV), indicating that it may provide a more independent estimate of phenotypic variability. Despite the removal of extreme outliers, a subset of lines retained unusually high variance estimates. While these observations may reflect genuine biological differences, they could also result from experimental sources of variation, such as developmental abnormalities or imperfect synchronization, which would tend to inflate estimates of intra-genotypic variance.

Phenotypic distribution

Our results show that variance in body size differs among *C. elegans* isogenic lines, thus suggesting there is a genetic effect on intra-genotypic variance. Specifically, we demonstrate that in the same environment, genetically identical individuals show different levels of among individual intra-genotypic variance depending on the genotype. There are genotypes with low intra-genotypic variance and others with high intra-genotypic variance. We therefore can conclude that this intra-genotypic variance is a property of a genotype. Furthermore, we observed that this intra-genotypic variance differed among genotypes suggesting that this variance may be due to the genotypes' different evolutionary histories.

As expected, worms exposed to the stressful environment were smaller than those maintained under standard laboratory conditions. This reduction in body size is consistent with physiological responses to osmotic stress, which can divert resources away from growth toward mechanisms involved in stress resistance and homeostasis [20]. More surprisingly, the effect of environment on intra-genotypic variance depended on the metric considered. Using $\ln(\sigma^2)$, worms exposed to stress exhibited lower intra-genotypic variance than those in the laboratory environment, whereas the opposite pattern was observed for ln(CV), with higher intra-genotypic variance under stressful conditions. These contrasting results likely reflect the different properties of the two variance metrics and their sensitivity to changes in trait means. From the perspective of phenotypic canalization, the reduction in intra-genotypic variance observed for $\ln(\sigma^2)$ could indicate tighter developmental regulation under stress, whereby only a restricted range of phenotypes can be successfully produced. However, we initially expected stressful conditions to increase intra-genotypic variance, as environmental challenges can favour a bet-hedging strategy in which individuals sharing the same genotype express a broader range of phenotypes. Such increased variance could enhance the probability that some individuals are closer to the fitness optimum in a novel or stressful environment. This prediction is more consistent with the pattern observed for ln(CV), suggesting that the inference regarding environmental effects on intra-genotypic variance may depend critically on how variance is quantified.

The weak correspondence in variance across environments reveals that intra-genotypic variance is highly context-dependent. Such environmental sensitivity suggests that intra-genotypic variance has a plastic component, with the same genotype expressing different levels of variability depending on environmental conditions [33]. This support the idea of micro-environmental plasticity in the genetic architecture underlying intra-genotypic variance.

Our findings highlight that intra-genotypic variance is not an intrinsic property of genotype alone, but it is also strongly modulated by environmental conditions.

Heritability

The broad sense heritability for intra-genotypic mean and variance might in principle be explained by additive genetic variance expected among fully inbred lines relative to the outbred populations from which they were derived. The broad sense heritability were low for the mean and the variance.

Interestingly, broad-sense heritability of the mean phenotype was higher under standard laboratory conditions than under salt stress. This pattern suggests that environmental stress increases the contribution of non-genetic sources of variation to trait expression, thereby reducing the proportion of intra-genotypic variance attributable to genetic differences among lines. In contrast, broad-sense heritability was higher for both variance metrics in the stressful environment. This result indicates that genetic differences among lines became more important in determining the extent of intra-genotypic variance under stress.

One possible explanation is that stressful conditions reveal genetic variation that remains largely hidden under laboratory conditions. In a stable environment, developmental processes may be highly buffered, reducing differences among genotypes in their ability to generate intra-genotypic variance. Under stress, however, this buffering capacity may break down, exposing genetic differences in developmental stability and increasing the genetic contribution to intra-genotypic variance. Such a pattern is consistent with the release of cryptic genetic variation, whereby environmental perturbations uncover genetic effects that are otherwise masked under optimal conditions.

More generally, these results suggest that while environmental stress reduces the genetic control of trait means, it may simultaneously increase the genetic control of variance itself. This supports the view that intra-genotypic variance is a genetically variable trait whose expression depends strongly on environmental context.

GWAS

We have shown that there is genetic variation in intra-genotypic mean and variance for body size in *C. elegans*. We assessed genetic variation separately for low-salt environment and for the high-salt environment, which enabled us to assess whether the genetic architecture of intra-genotypic mean and variance is different in both environments, or is common to multiple environments.

A large number of SNPs were associated with mean trait variation under standard laboratory conditions, whereas only a limited number of associations were detected under salt stress. Several SNPs were shared between the two environments, suggesting a partially conserved genetic architecture of the mean phenotype. However, the presence of environment-specific SNPs indicates a plastic genetic architecture, whereby distinct loci contribute to trait variation depending on environmental context. This pattern is plausible given that body size is a highly polygenic trait influenced by multiple developmental and physiological pathways, which may be differentially regulated un-

der salt stress, leading to the recruitment of additional or alternative genetic effects.

In contrast, the genetic architecture of intra-genotypic variance was primarily detectable under stressful conditions, which is consistent with our expectations. In the domesticated laboratory environment, worms are likely closer to an adaptive optimum, reducing the pressure for increased phenotypic dispersion. Conversely, under salt stress, increased phenotypic variability may be advantageous under a bet-hedging strategy, where a broader distribution of phenotypes increases the probability of matching a shifted or uncertain fitness optimum. In this context, the emergence of variance-associated loci with high effect in the stress environment is coherent.

Our results also revealed a limited set of shared SNPs between ln(CV) and ln(σ^2), consistent with the partial overlap expected between these two measures of phenotypic variability. It is worth noting that our analyses were performed using a relatively stringent false discovery rate threshold (FDR = 0.05), which may have reduced the number of detectable associations. Indeed, a more permissive exploratory threshold (FDR = 0.2) would reveal additional mean and variance associated signals for both conditions, suggesting that some genetic effects may be of smaller magnitude or context-dependent.

Future analyses on top SNPs may help refine the identification of independent signals, potentially improving the detection of true variance quantitative trait loci (vQTLs) and facilitating their comparison with previously characterized loci.

These findings indicate that variance can evolve independently of the mean and may itself constitute an evolvable trait [34]. The trajectories of response to natural or artificial selection for the mean and intra-genotypic variance depend on whether or not the same variants affecting the mean also affect intra-genotypic variance.

Conclusion

Overall, our results demonstrate that intra-genotypic variance is not merely a by-product of trait expression but a heritable and environmentally responsive trait in its own right. The weak correlations in variance across environments reveal a substantial plastic component, while the non-zero heritability estimates indicate that genetic factors contribute to differences in intra-genotypic variance. Furthermore, the partial overlap between loci associated with the mean and those associated

with intra-genotypic variance suggests that these two trait components share some, but not all, of their genetic architecture. Consequently, intra-genotypic variance may evolve both independently from and jointly with the mean, depending on the extent of their underlying genetic overlap. Understanding this relationship is essential for predicting evolutionary responses to natural or artificial selection, as selection acting on trait means may simultaneously alter intra-genotypic variance, whereas variance-specific loci may allow intra-genotypic variance itself to evolve as an independent target of selection.

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